

Graphing Basic Polar Curves from Equations

Answer Key

Precalculus

Quick Answers

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|--------------------|-----------------|-------------|
| 1. 3 | 5. -2 | 9. 120, 240 |
| 2. 2 | 6. 180 | 10. 0 |
| 3. 1 | 7. 30, 150, 270 | 11. — |
| 4. 0, 90, 180, 270 | 8. 4 | 12. — |
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Common wrong answers

5. *If they got 6: took cos of 180 degrees as +1 instead of -1, so the negative r never appeared*

1. $r = 6 \cos \theta$ at $\theta = 60^\circ$. Substitute the angle and use the exact cosine value: $\cos 60^\circ = \frac{1}{2}$, so $r = 6 \cdot \frac{1}{2}$. $\boxed{r = 3}$

2. $r = 4 \sin \theta$ at $\theta = 30^\circ$. Since $\sin 30^\circ = \frac{1}{2}$, $r = 4 \cdot \frac{1}{2}$. $\boxed{r = 2}$

3. $r = 1 + \cos \theta$ at $\theta = 90^\circ$. Here $\cos 90^\circ = 0$, so $r = 1 + 0$. The cardioid therefore reaches one unit straight up the 90° ray. $\boxed{r = 1}$

4. A rose returns to the pole exactly where $r = 0$:

$$4 \sin(2\theta) = 0 \implies \sin(2\theta) = 0$$

Sine vanishes at multiples of 180° , so $2\theta = 0^\circ, 180^\circ, 360^\circ, 540^\circ$ — the doubled window is what produces four answers, not two. Halving each gives

$$\boxed{\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ}$$

Between consecutive zeros the curve traces one petal, which is why $r = 4 \sin(2\theta)$ has four petals rather than two.

5. $r = 2 + 4 \cos \theta$ at $\theta = 180^\circ$. Since $\cos 180^\circ = -1$, $r = 2 + 4(-1) = 2 - 4$. $\boxed{r = -2}$

A negative r is not an error: the point is plotted 2 units from the pole in the direction *opposite* 180° , i.e. along the 0° ray. A student who takes $\cos 180^\circ$ as $+1$ gets 6 and misses the inner loop entirely.

6. Set the cardioid's radius to zero:

$$1 + \cos \theta = 0 \implies \cos \theta = -1$$

On $0^\circ \leq \theta < 360^\circ$ the cosine equals -1 at exactly one angle. $\boxed{\theta = 180^\circ}$ This lone zero is the cardioid's cusp, which is what distinguishes a cardioid from a limaçon with an inner loop (two zeros) or one without (none).

7. The petal tips are where the radius attains its maximum 2:

$$2 \sin(3\theta) = 2 \implies \sin(3\theta) = 1$$

Sine equals 1 at 90° plus multiples of 360° , so $3\theta = 90^\circ, 450^\circ, 810^\circ$ (the next value, 1170° , would put θ outside the window). Dividing by 3:

$$\boxed{\theta = 30^\circ, 150^\circ, 270^\circ}$$

Three tips, 120° apart — the three-petal rose you expect from an odd coefficient.

8. $r = 3 - 2 \cos \theta$ at $\theta = 120^\circ$. Since $\cos 120^\circ = -\frac{1}{2}$, $r = 3 - 2(-\frac{1}{2}) = 3 + 1$. Subtracting a negative increases r , which is why this limaçon bulges toward 180° . $\boxed{r = 4}$

9. The inner loop begins and ends where the curve passes through the pole:

$$2 + 4 \cos \theta = 0 \implies \cos \theta = -\frac{1}{2}$$

Cosine is $-\frac{1}{2}$ in the second and third quadrants, at

$$\boxed{\theta = 120^\circ, 240^\circ}$$

Between those two angles r is negative, and those points plot on the opposite side of the pole — that is the inner loop.

10. The circle's maximum radius occurs where

$$4 \cos \theta = 4 \implies \cos \theta = 1$$

On $0^\circ \leq \theta < 360^\circ$ this happens once. $\boxed{\theta = 0^\circ}$

So $r = 4 \cos \theta$ is the circle of diameter 4 sitting on the polar axis, with centre $(2, 0^\circ)$ — the maximum- r direction is exactly the direction of the offset.

11. Manual review — hand-drawn sketch. A correct graph shows:

- The four quadrantal points: $r = 2$ at $\theta = 0^\circ$, $r = 1$ at 90° , $r = 0$ at 180° , $r = 1$ at 270° .
- A smooth heart shape symmetric about the horizontal polar axis (because $\cos$ is even, $r(-\theta) = r(\theta)$).
- The cusp labelled at the pole, reached along the 180° direction.
- No part of the curve outside the ring $r = 2$, and no negative r : the radius $1 + \cos \theta$ never drops below 0.

12. Manual review — identification, sketch, and justification. A complete answer contains:

- *Family:* a limaçon of the form $r = a + b \cos \theta$, specifically a limaçon with an inner loop.
- *Test:* compare a with b . Here $a = 2$ and $b = 4$, so $a/b = 1/2 < 1$; the radius must change sign, and a polar curve crosses the pole exactly where it does. For $r = 4 + 2 \cos \theta$ the ratio is $2 > 1$, the radius stays positive, and no loop can form.
- *Evidence:* the pole crossings found in problem 9 are $\theta = 120^\circ$ and 240° ; between them $r < 0$ (for instance $r = -2$ at 180° , from problem 5), and those negative radii plot back through the pole to form the loop.
- *Sketch:* outer curve reaching $r = 6$ at $\theta = 0^\circ$, passing through the pole at 120° and 240° , with the small loop lying on the 0° side of the pole and extending to a distance 2. Symmetric about the polar axis.

Accept any justification that compares the constant with the coefficient and links the sign change of r to the pole crossings.